

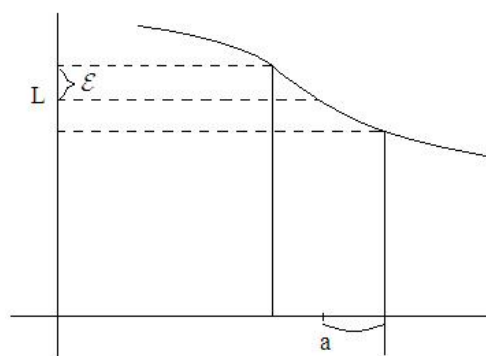
§2-4 Definition of the Limit

Homework : 3,5,13,15,31

Definition : $\lim_{x \rightarrow a} f(x) = L \Leftrightarrow$ Given $\varepsilon > 0$, there exists $(\exists) \delta > 0$ such that

(s.t.)

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad 0 < |x - a| < \delta.$$



Remarks : (i) ε 是任意給，可多小就有多小。

(ii) $|x - a| > 0 \Rightarrow x \neq a.$

(iii) 給 L 的任一小的近旁（附近），我們需找一個 a 的夠小近旁，使得在此找到的 a 的近旁，除 a 之外，每個相對的 f (or y) 值皆介於事先任意指定的 L 的近旁。

Example 1 : $\lim_{x \rightarrow 1} (5x - 1) = 4$

Proof by definition.

Proof :

草稿 :

Wanted :

$$|f(x) - L| < \varepsilon \quad (L \text{ 的近旁})$$

$$|5x - 1 - 4| < \varepsilon$$

$$\Rightarrow |5x - 5| < \varepsilon$$

$$\Rightarrow |x - 1| < \frac{\varepsilon}{5} \quad (a \text{ 的近旁})$$

Given ε_0 , chose $\delta = \frac{\varepsilon}{5}$. We have that

if $0 < |x-1| < \frac{\varepsilon}{5}$, then

$|5x-5| < \varepsilon$, that is

Example 2 : Prove $\lim_{x \rightarrow 3} x^2 = 9$ by definition.

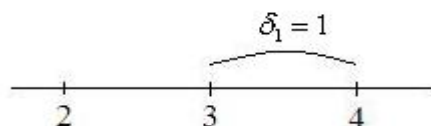
Proof :

草稿 :

Wanted :

$$|x^2 - 9| < \varepsilon$$

$$\Rightarrow |(x-3)(x+3)| < \varepsilon$$



若 $2 < x < 4$, 也即 $|x-3| < 1$,

$$\text{則 } |(x-3)(x+3)| < 7|x-3|$$

若 $|x-3|$ 也小於 $\frac{\varepsilon}{7}$, 則 $|(x-3)(x+3)| < 7|x-3| < \varepsilon$.

Given $\varepsilon > 0$, 取 $\delta = \min \left\{ 1, \frac{\varepsilon}{7} \right\} = 1$ 和 $\frac{\varepsilon}{7}$ 較小的那一數.

則 if $|x-3| < \delta$ then

$$|x^2 - 9| = |(x-3)(x+3)| < 7|x-3| < \varepsilon.$$